

Supply Chain Inventory Optimization Analysis

Executive Summary Report

Objective: Evaluate inventory health across products and suppliers to identify stock risks, operational bottlenecks, and opportunities for inventory optimization.

Executive Summary

Analysis of 100 products found that inventory levels were generally well balanced, with 90% classified as normal stock levels. However, 10 products were identified as either overstocked or understocked, indicating opportunities to improve inventory planning and operational efficiency.

Key Findings

1. Inventory Levels Are Largely Balanced

90 products were classified as Normal, 5 as Overstock Risk and 5 as Understock Risk.

2. Cosmetics Account for the Highest Concentration of Inventory Risk

The Cosmetics category contained the highest number of inventory risk flags.

3. Supplier Lead Times Alone Do Not Explain Inventory Risk

Inventory imbalances were not consistently associated with longer supplier lead times.

4. Understock Risk Is Associated With Longer Manufacturing Lead Times

Understocked products recorded the highest average manufacturing lead times (22.4 days).

5. Understocked Products Show The Highest Defect Rates

Understocked products recorded the highest defect rate (3.19%).

Recommendations

- Improve inventory planning and forecasting within the Cosmetics category.
- Investigate manufacturing bottlenecks.
- Strengthen quality control processes.
- Monitor products approaching inventory risk thresholds.

Conclusion

Overall inventory health is strong, with only 10% of products classified as inventory risks. Manufacturing delays and quality issues appear more closely linked to inventory risk than supplier lead times alone.